



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1 2025/2026

SECP3623 Database Programming

SECTION 02

Lab Assignment 2

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Question A1

```
> use clinicdb
< switched to db clinicdb
> db.createCollection("appointments")
< { ok: 1 }
> db.createCollection("loginSessions")
< { ok: 1 }
clinicdb> |
```

Question A2

Insert documents into appointments

```
> db.appointments.insertMany([
  {
    appointmentId: "A001",
    patientName: "Ali Hassan",
    doctor: "Dr. Teh",
    department: "Cardiology",
    status: "COMPLETED",
    fee: 180,
    appointmentDate: ISODate("2025-01-05")
  },
  {
    appointmentId: "A002",
    patientName: "Nur Aisyah",
    doctor: "Dr. Lim",
    department: "Dermatology",
    status: "PENDING",
    fee: 90,
    appointmentDate: ISODate("2025-01-10")
  },
])
```

```
{
  appointmentId: "A003",
  patientName: "Jason Wong",
  doctor: "Dr. Ahmad",
  department: "Orthopedics",
  status: "CANCELLED",
  fee: 120,
  appointmentDate: ISODate("2025-01-12")
},
{
  appointmentId: "A004",
  patientName: "Siti Aminah",
  doctor: "Dr. Kumar",
  department: "Neurology",
  status: "COMPLETED",
  fee: 250,
  appointmentDate: ISODate("2025-01-08")
},
```

```
{
  appointmentId: "A005",
  patientName: "Daniel Lee",
  doctor: "Dr. Teh",
  department: "Cardiology",
  status: "PENDING",
  fee: 160,
  appointmentDate: ISODate("2025-01-15")
},
{
  appointmentId: "A006",
  patientName: "Farah Zainal",
  doctor: "Dr. Lim",
  department: "Dermatology",
  status: "COMPLETED",
  fee: 100,
  appointmentDate: ISODate("2025-01-18")
}
])
```

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('696893397cc0d66afe2f1d64'),
    '1': ObjectId('696893397cc0d66afe2f1d65'),
    '2': ObjectId('696893397cc0d66afe2f1d66'),
    '3': ObjectId('696893397cc0d66afe2f1d67'),
    '4': ObjectId('696893397cc0d66afe2f1d68'),
    '5': ObjectId('696893397cc0d66afe2f1d69')
  }
}
clinicdb >
```

Insert documents into loginSessions

```
> db.loginSessions.insertMany([
  {
    staffId: "S001",
    staffName: "Ahmad Firdaus",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "192.168.1.10",
    loginTime: ISODate("2025-01-15T08:30:00")
  },
  {
    staffId: "S002",
    staffName: "Siti Noor",
    role: "Nurse",
    status: "ACTIVE",
    ipAddress: "192.168.1.12",
    loginTime: ISODate("2025-01-15T09:00:00")
  },
])
```

```
{
  staffId: "S003",
  staffName: "Jason Lim",
  role: "Receptionist",
  status: "INACTIVE",
  ipAddress: "192.168.1.15",
  loginTime: ISODate("2025-01-14T17:45:00")
},
{
  staffId: "S004",
  staffName: "Nur Huda",
  role: "Admin",
  status: "ACTIVE",
  ipAddress: "192.168.1.20",
  loginTime: ISODate("2025-01-15T07:50:00")
},
```

```

{
  staffId: "S005",
  staffName: "Daniel Wong",
  role: "Doctor",
  status: "INACTIVE",
  ipAddress: "192.168.1.22",
  loginTime: ISODate("2025-01-13T18:10:00")
},
{
  staffId: "S006",
  staffName: "Aina Rahman",
  role: "Nurse",
  status: "ACTIVE",
  ipAddress: "192.168.1.25",
  loginTime: ISODate("2025-01-15T10:15:00")
}
])

```

```

< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('696899947cc0d66afe2f1d6d'),
    '1': ObjectId('696899947cc0d66afe2f1d6e'),
    '2': ObjectId('696899947cc0d66afe2f1d6f'),
    '3': ObjectId('696899947cc0d66afe2f1d70'),
    '4': ObjectId('696899947cc0d66afe2f1d71'),
    '5': ObjectId('696899947cc0d66afe2f1d72')
  }
}
clinicdb>

```

Question B1

```
> db.appointments.find(  
  {  
    status:"COMPLETED",  
    fee:{$gte: 100, $lte: 300}  
  },  
  {  
    _id: 0,  
    appointmentId: 1,  
    patientName: 1,  
    department: 1,  
    fee: 1}).sort(  
  {  
    department: 1,  
    fee: -1  
  })
```

```
< {  
  appointmentId: 'A001',  
  patientName: 'Ali Hassan',  
  department: 'Cardiology',  
  fee: 180  
}  
  
{  
  appointmentId: 'A006',  
  patientName: 'Farah Zainal',  
  department: 'Dermatology',  
  fee: 100  
}  
  
{  
  appointmentId: 'A004',  
  patientName: 'Siti Aminah',  
  department: 'Neurology',  
  fee: 250  
}  
clinicdb >
```

Question B2

```
> db.appointments.aggregate([
  {
    $group:
    {
      _id:"$department",
      totalAppointments:{$sum:1},
      totalFees:{$sum:"$fee"}
    }
  }
]).sort({totalFees:-1})
< {
  _id: 'Cardiology',
  totalAppointments: 2,
  totalFees: 340
}
{
  _id: 'Neurology',
  totalAppointments: 1,
  totalFees: 250
}
```

```
{
  _id: 'Dermatology',
  totalAppointments: 2,
  totalFees: 190
}
{
  _id: 'Orthopedics',
  totalAppointments: 1,
  totalFees: 120
}
clinicdb>
```

Question C1

```
> db.loginSessions.find(
  {status:"ACTIVE"},
  {_id:0, staffName:1, role:1, ipAddress:1, loginTime:1}
).sort({loginTime:-1})
< {
  staffName: 'Aina Rahman',
  role: 'Nurse',
  ipAddress: '192.168.1.25',
  loginTime: 2025-01-15T10:15:00.000Z
}
{
  staffName: 'Siti Noor',
  role: 'Nurse',
  ipAddress: '192.168.1.12',
  loginTime: 2025-01-15T09:00:00.000Z
}
{
  staffName: 'Ahmad Firdaus',
  role: 'Doctor',
  ipAddress: '192.168.1.10',
  loginTime: 2025-01-15T08:30:00.000Z
}
```

```
{
  staffName: 'Nur Huda',
  role: 'Admin',
  ipAddress: '192.168.1.20',
  loginTime: 2025-01-15T07:50:00.000Z
}
clinicdb>
```

Question C2 (a)

```
> db.loginSessions.distinct("ipAddress",{status: "ACTIVE"})  
< [ '192.168.1.10', '192.168.1.12', '192.168.1.20', '192.168.1.25' ]
```

Question C2 (b)

```
> db.loginSessions.distinct("staffId", {status: "ACTIVE"}).length  
< 4
```

Question D1

```
> db.appointments.createIndex({department:1, status: 1})  
< department_1_status_1
```

Question D2 (a)

```
> db.appointments.updateMany(  
  {status:"CANCELLED"},  
  {$set: {status:"ARCHIVED"}}  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

Question D2 (b)

```
> db.loginSessions.deleteMany({status: "INACTIVE"})
< {
  acknowledged: true,
  deletedCount: 2
}
clinicdb>
```